# Influence Lines

## Lab S2 – Cantilever (Gerber) Bridge Model

SEAA 3012 Civil Engineering Laboratory II

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Faculty of Civil Engineering, UTM

# Today

1. Why influence lines? What they are, what they are not
2. The laboratory model and why it is determinate
3. Deriving the six influence lines from statics
4. The Müller–Breslau shortcut
5. Using an influence line: one load, a train of loads, a distributed load
6. The experiment: data reduction, errors, what to report



The bridge model in the Structures Laboratory.

# Why influence lines?

# Loads that move

A floor load stays put. A lorry on a bridge, a crane on its rail, a train on a viaduct do not.

The designer needs to know where the load must stand to produce

- the largest reaction at a pier (bearing design, uplift),
- the largest shear at a section,
- the largest moment at midspan.

A shear-force or bending-moment diagram cannot answer this: it assumes the loads are fixed.

**Definition**

The **influence line** of a response $R$ at a fixed point is the graph of $R$ against the position $x$ of a **unit** load moving over the structure: ordinate $\eta_R(x)$, units of $R$ per unit load.

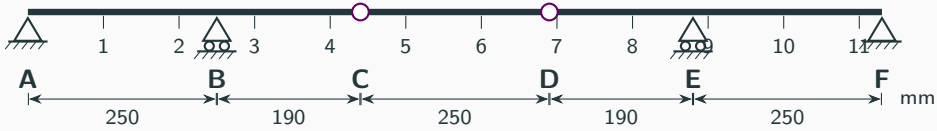
# Influence line versus bending-moment diagram

| | SFD / BMD | Influence line |
|---|---|---|
| Fixed | the loads | the point where $R$ is read |
| Varies along $x$ | the section where $V$, $M$ are read | the position of the unit load |
| Ordinate means | $V$ or $M$ at that section | $R$ at the fixed point for a load at $x$ |

Because the structure is linear elastic, effects add: once $\eta_R(x)$ is known for a unit load, any load pattern is handled by multiplication and addition.

# The laboratory model

# A Gerber (cantilever) bridge, 1130 mm long




- Beam ABC: pin A, roller B, cantilever BC (190 mm)
- Suspended span CD (250 mm) rests on hinges at C and D
- Beam DEF: mirror image of ABC
- Gauges under A, B, E, F read the support reactions
- Gauges under C, D read the hinge forces (span pressing on the tips)
- Halts 1–11: where the mass is placed

# Why it is statically determinate

**Unknowns:** $R_A$, $R_B$, $R_E$, $R_F$ (no horizontal load, so no horizontal reactions) – four.

**Equations:**

- $\sum F_y = 0$ and $\sum M = 0$ for the whole model – two
- $M = 0$ at hinge C and at hinge D – two more

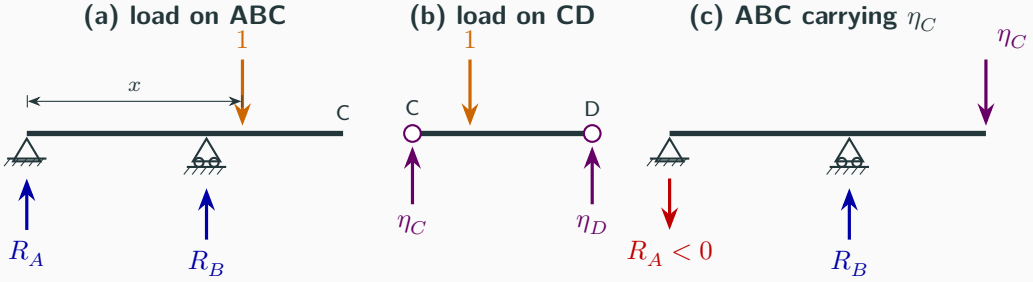
Four equations, four unknowns.

**Consequences**

- Every reaction follows from equilibrium; $EI$ never appears.
- Every influence line is made of straight segments, with kinks only at the hinges.
- Sign convention: reactions positive *up*; $R_C$, $R_D$ positive when the span presses *down* on the tip.

# Deriving the influence lines

# One idea, used three times: take moments about a support




**Order matters:** the suspended span is analysed first, because it is the only member whose loads reach the others.

## Step 1 – load on beam ABC ($0 \le x \le 440$)

Moments about B for beam ABC (hinge C unloaded):

$$R_A \cdot 250 - 1 \cdot (250 - x) = 0 \quad \Longrightarrow \quad \eta_A(x) = \frac{250 - x}{250}, \qquad \eta_B(x) = 1 - \eta_A = \frac{x}{250}.$$

- Nothing stops at B: for $250 < x \le 440$ the lever arm $(250 - x)$ is negative
- So $\eta_A < 0$: support A must *hold the beam down*
- and $\eta_B > 1$: B carries the load *and* the pull-down at A

**At the tip C**

$\eta_A(440) = -190/250 = \mathbf{-0.76}$
$\eta_B(440) = 440/250 = \mathbf{1.76}$

## Steps 2 and 3 – load on the suspended span ($440 \le x \le 690$)

**Step 2.** CD is a simply supported beam on the hinges (lever rule):

$$\eta_C(x) = \frac{690 - x}{250}, \qquad \eta_D(x) = \frac{x - 440}{250}, \qquad \eta_C + \eta_D = 1.$$

**Step 3.** Beam ABC now carries only $\eta_C$ at its tip, i.e. Step 1 with the load standing at $x = 440$:

$$\eta_A(x) = -0.76\,\eta_C(x), \qquad \eta_B(x) = 1.76\,\eta_C(x).$$

- At C both formulae give $\eta_A = -0.76$: the line is continuous
- At D, $\eta_C = 0$, hence $\eta_A = \eta_B = 0$: a load at D is carried entirely by DEF

# Step 4 – load on beam DEF, and the checks

DEF is the mirror image of ABC (E $\leftrightarrow$ B, F $\leftrightarrow$ A):

$$\eta_F(x) = \frac{x - 880}{250}, \quad \eta_E(x) = \frac{1130 - x}{250}$$

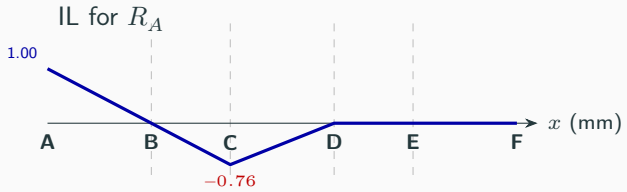
for $690 \le x \le 1130$; for a load on CD, $\eta_E = 1.76\,\eta_D$ and $\eta_F = -0.76\,\eta_D$.

A load on ABC never reaches DEF (and vice versa): the suspended span is unloaded and transmits nothing.

**Three checks**

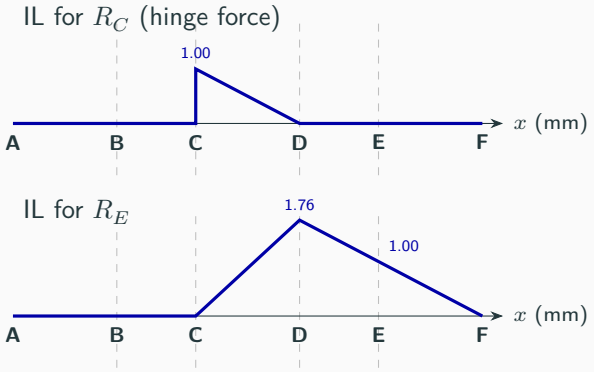
1. $\eta_A + \eta_B + \eta_E + \eta_F = 1$ for every $x$
2. Continuity at the hinges C and D
3. Slopes: $-1/250$ per mm on ABC, $+0.76/250$ on CD; kinks only at hinges, *never* at B

# The influence lines (1): the support reactions $R_A$ and $R_B$




| Halt | $x$ | $\eta_A$ | $\eta_B$ |
|---|---|---|---|
| A | 0 | 1.00 | 0.00 |
| 1 | 100 | 0.60 | 0.40 |
| 2 | 200 | 0.20 | 0.80 |
| B | 250 | 0.00 | 1.00 |
| 3 | 300 | $-0.20$ | 1.20 |
| 4 | 400 | $-0.60$ | 1.60 |
| C | 440 | $-0.76$ | 1.76 |
| 5 | 500 | $-0.58$ | 1.34 |
| 6 | 600 | $-0.27$ | 0.63 |
| D | 690 | 0.00 | 0.00 |

# The influence lines (2): hinge force $R_C$, and the right-hand side



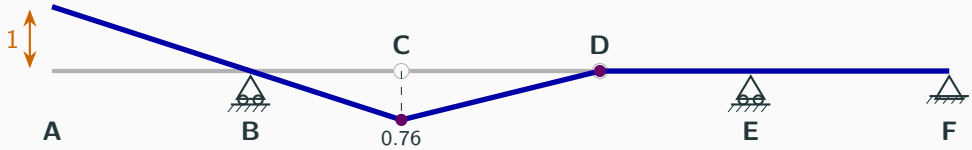

- $R_C$ is non-zero only for a load on the suspended span: a triangle, 1 at C, 0 at D
- $R_D$ is its mirror image (0 at C, 1 at D)
- $R_E$ and $R_F$ are the mirror images of $R_B$ and $R_A$ about $x = 565$ mm
- Reactions at B and E exceed the load (1.76) when it stands on the cantilever tips

# The Müller–Breslau principle

# The shortcut practising engineers use

**Müller–Breslau (1886)**

Remove the restraint corresponding to $R$, impose a unit displacement in the direction of $R$: the **deflected shape** is the influence line of $R$.




For $R_A$: release A, lift it by 1. ABC rotates about B as a rigid bar (C drops $190/250 = 0.76$); CD follows C and stays straight; DEF does not move. Same line, same signs, no algebra.

## What the principle tells you at a glance

- Kinks only at hinges and at the released support – a kink at B means an error
- Where the structure moves *against* the released reaction, the ordinate is negative: uplift at A for a load between B and D
- Exact ordinates for *determinate* structures (rigid links)
- Only the *shape* for indeterminate ones: the released beam bends, the line is curved

**Try it (Exercise 5)**

Sketch $\eta_E$ and $\eta_F$ by releasing E and F. Which supports can ever see uplift, and for which load positions?

# Using an influence line

# Three rules

$R = P\,\eta_R(x_1)$ — single load $P$ at $x_1$

$R = \sum_i P_i\,\eta_R(x_i)$ — series of loads

$R = w \times$ (area under the IL from $a$ to $b$) — uniform load $w$ on $[a, b]$

**Critical position**

All our ILs are piecewise linear, so $\sum_i P_i\,\eta_R(x_i)$ is piecewise linear in the position of the group: its extreme occurs with one load on a vertex (A, B, C or D). Only a few positions need to be tried.

**Units**

With masses in kg the reactions come out in kgf, which is what the gauges read ($1$ div $= 0.05$ kgf).

# Worked example 1 – the compound load of Test 2

$2.5$ kg mass $100$ mm to the right of the $5$ kg mass (in front, moving right):
$R_A(x) = 5\,\eta_A(x) + 2.5\,\eta_A(x+100)$.

- 5 kg mass at A:
  $5(1.0) + 2.5(0.6) = \mathbf{6.5}$ kgf
- 5 kg mass at C:
  $5(-0.76) + 2.5(-0.456) = \mathbf{-4.94}$ kgf
- 2.5 kg mass at C (5 kg at 340):
  $5(-0.36) + 2.5(-0.76) = -3.70$ kgf

The most severe pull-down occurs with the heavier load on the vertex.

| 5 kg at | $x$ | $R_A$ (kgf) | $R_B$ (kgf) |
|---|---|---|---|
| A | 0 | 6.50 | 1.00 |
| 1 | 100 | 3.50 | 4.00 |
| 2 | 200 | 0.50 | 7.00 |
| B | 250 | −1.00 | 8.50 |
| 3 | 300 | −2.50 | 10.00 |
| 4 | 400 | −4.44 | 11.34 |
| C | 440 | −4.94 | **11.44** |
| 5 | 500 | −3.57 | 8.27 |
| 6 | 600 | −1.37 | 3.17 |

# Worked example 2 – a uniformly distributed load

Areas under $\eta_A$: $+125$ mm (AB), $-72.2$ mm (BC), $-95.0$ mm (CD), 0 (DEF).

For $w = 0.01$ kg/mm:

- over AB only: $R_A = 0.01 \times 125 = 1.25$ kgf (half the load on a simple span)
- over BC and CD: $R_A = 0.01 \times (-167.2) = -1.67$ kgf
- over the whole model: $R_A = 0.01 \times (-42.2) = -0.42$ kgf – the deck *lifts* at A

**Check by statics**

ABC carries $w(250 + 190)$ directly plus $125w$ at C. Moments about B:
$R_A \cdot 250 = 250w(125) - 190w(95) - 125w(190) = -10\,550w$, so $R_A = -42.2w$.
The IL gave it without a new free-body diagram.

# The experiment

# What you will do, and what the numbers mean

**Test 1:** $2.5\,\text{kg}$ mass at each of the $17$ halts; read all six gauges.
**Test 2:** $5\,\text{kg} + 2.5\,\text{kg}$ pair, $100\,\text{mm}$ apart.

## Data reduction

$$\eta_{\text{test}} = \frac{\text{reading (div)} \times 0.05}{m}, \qquad m = 2.5\,\text{kg}$$

Test 2: reading $\times\, 0.05$ is the reaction in kgf.

**Before leaving the lab**

- Equilibrium check at every halt: $(a + b + e + f) \times 0.05 \approx 2.5$ kgf
- Re-zero the gauges between halts
- Centre the mass on the halt mark ($\pm 5$ mm matters)

**In the report:** theoretical and experimental ILs on the same axes, per reaction; absolute differences at every halt.

# How good is “good agreement”?

| Source | Estimate | on $\eta_A$ |
|---|---|---|
| Mass $\pm 5$ mm off the halt | slope $\times\, 5$ mm | $\pm 0.020$ |
| Gauge $\pm 1$ division | $0.05/2.5$ | $\pm 0.020$ |
| Zero drift | re-zero each halt | $\pm 0.02$–$0.04$ |
| Friction in hinges, gauges | systematic | sum check |

Combined: about $\pm 0.03$ per unit load, the same at every halt.

**Percentages mislead**

$\pm 0.03$ is 3 % of the ordinate at A (1.0) but 15 % at halt 2 (0.2), and undefined where the ordinate is zero.

Compare by **absolute difference**; quote percentages only at A, 1, 4, C, 5.

# Activity and summary

## Activity: Influence Line Explorer

Scan the code. *Before each step*, write your prediction, then check it.

1. Move a unit load from A to F. Where does $R_A$ change sign? Where is $R_B$ largest, and why is it larger than the load?
2. Switch to the $5\,\text{kg} + 2.5\,\text{kg}$ pair; find by trial the most negative $R_A$ (expect $-4.94$ kgf).
3. Shade the areas under $\eta_A$: confirm $-42.2$ mm net.
4. Answer the six prediction questions.
5. In the lab: type your gauge readings into the last panel and compare with theory before writing the report.

Influence Line Explorer

# Summary

**Model: six straight-line influence lines**

$\eta_A = (250 - x)/250$ on ABC, $-0.76\,\eta_C$ on CD, 0 on DEF

$\eta_B = x/250$ on ABC, $1.76\,\eta_C$ on CD

$\eta_C = (690 - x)/250$,

$\eta_D = (x - 440)/250$ on CD

$\eta_E$, $\eta_F$: mirror images of $\eta_B$, $\eta_A$

Check: $\eta_A + \eta_B + \eta_E + \eta_F = 1$

**Use**

$R = \sum P_i\,\eta_R(x_i)$; UDL: $R = w \times$ area.

Extreme of a load train: one load on a vertex.

Müller–Breslau: release, displace by 1, read the shape.

**Report**

$\eta_{\text{test}} =$ reading $\times\, 0.05/m$; equilibrium check; absolute differences; percentages only at large ordinates.

Reading: Hibbeler, *Structural Analysis*, 10th ed., ch. 6; Kassimali, *Structural Analysis*, 6th ed., ch. 8; Laboratory Manual, experiment S2.